Frame It as Cases: Work That Speaks for Itself

Ahmed Elgharably — FlyRank ML Internship — General AI Fluency, Week 2

Voice Card

> **Direct, warm, evidence-first, no buzzwords, builds from the ground up.**

I write like I talk. I show my work. I don't claim things the data can't back. If I wouldn't say it out loud to someone I respect, it doesn't go in.

Case Study: Refresh / Content Opportunity Scoring

FlyRank ML Engineering Internship — Capstone Lane Selection & Research Framing

The Problem

A content team at a digital publisher has thousands of pages and maybe twenty hours a week to review them. Which pages deserve those twenty hours? Pick wrong, and two things happen: you waste time rewriting a page that was already fine, or you miss a page that's quietly bleeding traffic. The second one hurts more — a high-impression page in decline has a shrinking window to recover.

I was two weeks into the FlyRank ML internship when I hit this. The assignment wasn't "build a model." It was: pick a lane, frame a question, and prove with real numbers that the lane is worth seven more weeks of your life.

What I Did (and What I Decided)

I read the full lane guide — sixteen sections, from data contracts to leakage audits — before I touched a line of code. Most people skip to the model. I didn't. I needed to understand what "the right page to fix" actually means in this context: it's not a page guaranteed to recover. It's the page a reviewer should look at *first*, given limited capacity.

I chose **Lane 2: Refresh / Content Opportunity Scoring**. Not because it sounded impressive, but because it produces a ranked queue with reason codes — something a real person can act on Monday morning. The other lanes were either too exploratory (signal analysis) or less actionable without text data (clustering).

Then I pulled three real numbers from the starter dataset:

- **1,847 pages** are both visible (500+ impressions in 90 days) *and* stale (6+ months old). That's 6.2% of the inventory sitting there with an audience but aging content.
- **2,103 pages** are declining (trend = down) *and* have meaningful demand (100+ impressions). That's 7.0% losing visibility despite having search interest.
- **1,156 pages** rank on page 1–2 (position 1–20) but have CTR under 0.5%. That's 3.9% under-capturing clicks they should be getting.

These three segments alone gave me a clear, evidence-backed case that the data supports a refresh-scoring project. I didn't make up the thresholds — they came from the starter baseline rules, which I treated as a transparent starting point to beat, not a ceiling.

I also wrote down what I *cannot* claim. I cannot say refreshing a page causes recovery — that needs an experiment. I cannot say I'm predicting Google's algorithm — I only see observable signals. This honesty isn't modesty; it's what keeps the work trustworthy.

What Came of It

- A research question notebook that runs top-to-bottom with no errors, every claim tied to a real number from the data.
- A lane choice backed by evidence instead of gut feeling.
- A clear understanding that my job isn't to build magic SEO automation. It's to help a human reviewer spend their limited time on the pages where the data shows the strongest evidence of opportunity or decay.
- The starter baseline achieves Precision@50 of 0.24. The random forest model hits 0.74. That's a 3× lift — proof that learning from patterns beats fixed rules, but only if you're honest about what those patterns can and cannot say.

Bio

I'm Ahmed Elgharably, an ML Engineering intern at FlyRank. I work at the intersection of search data and human decisions — building ranked, evidence-backed tools that help teams with limited capacity focus on what actually matters. I believe the best ML work is the kind you can explain out loud, defend with real numbers, and hand to someone who needs it on a Monday morning.

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Before / After: Generic AI vs. My Voice

Generic AI line (what I *didn't* submit):

> "Leveraged cutting-edge machine learning algorithms and advanced data analytics to optimize content refresh strategies, driving measurable SEO performance improvements and enhancing digital discoverability across the client portfolio."

What's wrong with it: I would never say "leveraged" or "cutting-edge" or "driving measurable SEO performance improvements" out loud. It sounds like a press release written by a robot. It hides the actual decision. It claims "measurable improvements" when I haven't run the experiment yet. It makes me sound like I'm selling something.

My edited version:

> "I built a ranked queue that helps a reviewer with twenty pages a week of capacity pick the right ones first. I used search and engagement signals the data actually shows — impressions, clicks, content age, trend direction — and I was honest about what I can claim (this page looks like others that declined) and what I can't (refreshing it will definitely work)."

Why it's better: It names the real person (the reviewer), the real constraint (twenty pages), and the real uncertainty. It uses words I would actually say. It doesn't pretend the model is magic. It tells the truth.

Submitted to the General AI Fluency track thread.